Stability and Control Design Plan for a Titan Sea Drone

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I. Introduction and Mission Statement

The objective of this design project is to develop a conceptual aerial drone capable of operating in Titan's atmosphere and landing on its seas. For context, Titan is the largest moon of Saturn and has a dense, cryogenic atmosphere and hydrocarbon seas [1]. Battery-powered flight will be used by the proposed flight vehicle. This battery-powered flight will be supported by radioisotope thermoelectric generator (RTG) trickle charging. Its preliminary mission requirement is to fly at least 10 kilometers per battery charge, land on a Titan sea and remain afloat, and collect samples using a vacuum for compositional and organic analysis. This concept essentially combines the mobility of a rotorcraft and the ability to land in water of a seaplane.

The preliminary mission profile consists of the following: system activation and rotor deployment, vertical takeoff, hover, forward flight toward a selected sampling location, controlled descent, water touchdown, and floating operations while the scientific instruments collect samples. The flight vehicle must maintain stability and controllability during each phase of its flight. It must also maintain adequate flotation stability after landing. Its ability to re-take off or continue to different sampling locations will be evaluated as the mission concept continues to develop.

Major design constraints include the following: extremely low operating temperature, limited stored battery energy, aeroshell packaging requirements, rotor performance in Titan's atmosphere, scientific-payload mass, and the need for external mechanisms to function under cryogenic conditions. The project will initially use conceptual and parameterized analyses due to a final flight vehicle configuration not having been selected yet. A demonstrator will be developed based on Earth as well to evaluate the flight computer, rotor and propeller performance, water-landing and sampling methods, and the integration of the major subsystems.

Stability and control are essential to the mission because controlled flight must be maintained without exceeding its own available thrust, power, and actuator limits. Therefore, the stability and control design of the mission vehicle will influence rotor arrangement, center of gravity limits, sensor selection, actuator requirements, and overall vehicle configuration.

II. Individual Scope of Work

As the Stability and Control Lead, I will be responsible for the Sea Drone's flight dynamics and control architecture development and evaluation. My work in the project will include the following: stability and controllability requirements for hover, forward flight, descent, water landing, and possible takeoff from the sea, if decided on. Key performance indicators, including attitude-tracking error, settling time, damping ratio, available control authority, allowable center of gravity range, and actuator saturation limits, will also be established by me.

MATLAB will be used to develop a parameterized flight-dynamics model. Simulink will be used as the design matures and continues to develop as well. The initial model will represent the flight vehicle when it is near hover. This initial model will later be expanded upon to include forward flight and descent. The model will incorporate the following properties: vehicle mass, center of gravity location, moments of inertia, rotor locations, rotor thrust, rotor torque, and actuator response. These listed properties have not been decided on yet; therefore, until they are specified, they will be represented as variable inputs.

The flight vehicle's equations of motion (EOM) will be linearized about selected operating conditions so that state-space methods can be applied to the drone. The flight vehicle's dynamic modes will be identified using eigenvalues and eigenvectors. Whether or not the proposed rotor arrangement can independently control altitude, roll, pitch, and yaw will be determined by controllability analysis. Candidate feedback control options, such as proportional-integral-derivative control and linear-quadratic regulation, will be compared according to the following: stability, tracking performance, robustness, implementation difficulty, and computational requirements.

I will require coordination with the other subsystem leads for my stability and control work. Mass properties and rotor placement information will be provided by the Structures Lead. Rotor thrust, torque, motor response, and power-limit information will be provided by the Propulsion and Power Lead. Aerodynamic force, moment, and rotor-performance information will be provided by the Aerodynamics Lead. The flight computer, sensor, and actuator specifications will be provided by the Electronics Lead.

III. Methods and Approaches

The stability and control analysis is planned to begin with a parameterized MATLAB model because the flight vehicle is still in the conceptual design stage of the project. This approach will allow the model to be updated and improved upon as the team decides on vehicle geometry and components. The required inputs for the model will include the following: mass, center of gravity location, moments of inertia, rotor count and placement, rotor spin directions, thrust and torque coefficients, motor response time, and Titan atmospheric properties. Each of these listed inputs will be recorded as either known, estimated, or to be determined (TBD).

Hover will be the first operating condition to be analyzed. At hover, the total thrust of the rotor must equal the weight of the vehicle as shown in Eq. (1):

$$\sum_{i=1}^{N} T_i = mg \tag{1}$$

The thrust of the rotor and reaction torque will initially be modeled as shown in Eq. (2) and Eq. (3):

$$T_i = k_T \Omega_i^2 \tag{2}$$

$$Q_i = k_Q \Omega_i^2 \tag{3}$$

In Eq. 2 and Eq. 3, k_T and k_Q are rotor coefficients, and Ω_i is the speed of the rotor. The motion of the flight vehicle will be represented by the rigid-body force and moment equations as shown in Eq. (4) and Eq. (5):

$$m\,(\dot{\boldsymbol{v}} + \boldsymbol{\omega} \times \boldsymbol{v}) = \boldsymbol{F} \tag{4}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \boldsymbol{M} \tag{5}$$

After the team decides on a rotor arrangement, individual rotor commands will be related to total thrust and roll, pitch, and yaw moments by a control-allocation matrix. This analysis will determine whether the flight vehicle has enough control authority. It will also help establish allowable center of gravity limits.

A state-space model will be obtained by linearizing the nonlinear equations about hover as shown in Eq. (6):

$$\delta\dot{\boldsymbol{x}} = \mathbf{A}\delta\boldsymbol{x} + \mathbf{B}\delta\boldsymbol{u} \tag{6}$$

MATLAB will be used to calculate the following: eigenvalues, eigenvectors, damping ratios, and controllability. The eigenvalues will indicate whether each motion decays, grows, or remains constant; on the other hand, the eigenvectors will identify the states involved in each mode. The proposed arrangement of the rotor will be determined to assess whether it can independently control altitude, roll, pitch, and yaw through controllability analysis. Hover thrust per rotor, thrust margin, maximum control moments, angular acceleration, and allowable center of gravity range will be estimated using additional calculations. The model will be expanded later to represent forward flight, the planned descent profile, and the approach to the hydrocarbon sea landing.

After the preliminary flight vehicle model is established, a feedback controller will be added. Proportional-integral-derivative control and linear-quadratic regulation will be compared to each other using the following: tracking error, settling time, overshoot, damping, robustness, implementation difficulty, and actuator saturation. Feedback loops, motor response, sensor behavior, and disturbances may eventually be represented using Simulink. The proposed guidance, navigation, and control architecture will be updated and built upon previous Titan rotorcraft research, such as Dragonfly [2].

Candidate sensors will include the following: an inertial measurement unit, an altitude or range sensor, and navigation cameras. Flight-computer capabilities, sensor accuracy, update rates, and actuator specifications will be provided by the Electronics Lead. The sensors for the Earth demonstrator and the sensors proposed for Titan will be distinguished. Trade studies will compare the following to each other as evaluation criteria: rotor arrangements, sensing options, actuator methods, and control algorithms, using stability, control authority, power consumption, mass, reliability, aeroshell compatibility, and cryogenic performance.

The verification process will start with simulation and progress to physical testing. Planned activities include the following: MATLAB and Simulink analysis, motor and rotor characterization, sensor calibration, restrained flight testing, low-altitude hover testing, and free-flight testing. Separate sets of parameters for Earth and Titan will be maintained. This is because Titan performance cannot be directly proven by the Earth flight test; however, the Earth flight test will validate the implementation. Test results and model predictions will be compared to each other to update parameters that are still uncertain. The Titan sea-landing analysis will also establish an acceptable descent rate and touchdown attitude. This will be done in coordination with the Structures Lead and other team leads responsible for the flight vehicle's flotation.

Quantitative stability and control results have not been obtained this early in the design process yet. However, preliminary planning as a team has identified some vehicle mass properties, rotor geometry, and actuator characteristics as key inputs that still need to be determined. This means the initial model will therefore use estimated parameters until the analysis is updated when the team decides on a baseline configuration.

Mission success criteria will include the following: stable closed-loop eigenvalues, acceptable damping and settling time, satisfactory control margin, full controllability of the required flight states, and operation without sustained actuator saturation. Once the team selects a baseline configuration, numerical limits will be established. The immediate outputs of all this work will be a control-system block diagram, a table of assumptions, a trade study matrix, and a parameterized hover model of the flight vehicle.

IV. Project Timeline, Risk Management, and Milestones

The stability and control schedule will align with the team's major milestones and important phases of the Titan Sea Drone's development. The requirements and subsystem interfaces for the design project will be specified in the Mission Concept Review (MCR) on September 25, 2026. The parameterized hover model and the control analysis of the drone will support the Preliminary Design Review (PDR) on October 28, 2026. The team proposal on November 20, 2026, will be supported by the controller trade studies and the verification plan. This progress will be documented in the Individual Report on December 9, 2026. Work will continue in the Spring and will progress through controller refinement, subsystem integration, and staged ground and flight testing. The Fall stability and control Gantt chart is shown in Fig. 1. The Spring stability and control Gantt chart is shown in Fig. 2.

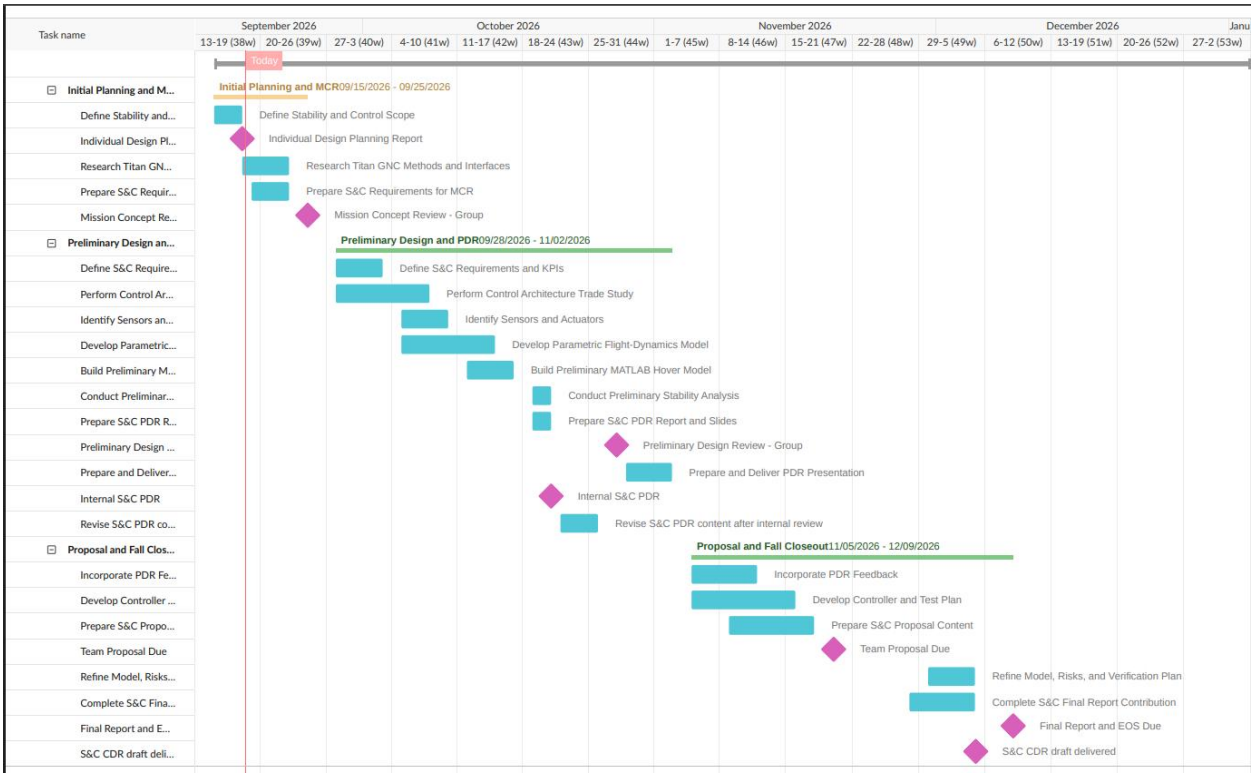


Fig. 1. S&C Fall Gantt Chart

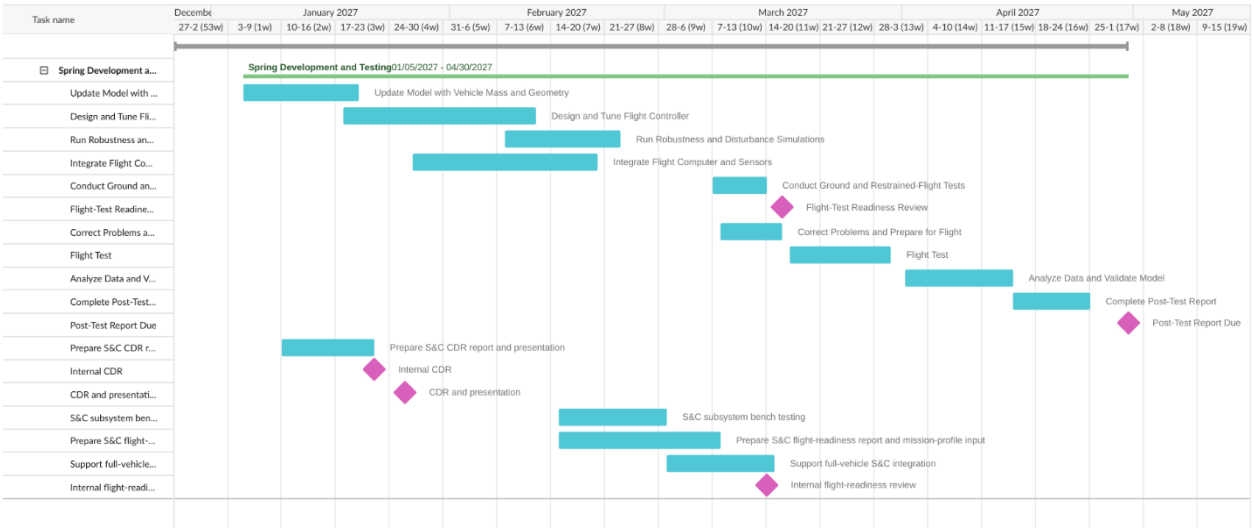


Fig. 2. S&C Spring Gantt Chart

The preliminary risk assessment ratings that will be reviewed by the team as the design develops are shown in Table 1.

Table 1. Risk Assessment Matrix

Risk	Likelihood	Impact	Mitigation
Uncertain mass properties and rotor data	High	Medium	Evaluate estimated parameter ranges and update the model as soon as subsystem data becomes available.
Insufficient control authority or actuator saturation	Medium	High	Check thrust margins and center of gravity limits early; coordinate changes to rotor geometry or actuator sizing.
Sensor and actuator integration delays	Medium	High	Confirm interfaces early; test components before full integration.
Inaccurate prediction of Titan performance	Medium	High	Maintain separate Earth and Titan models; document the limits of Earth testing.

All progress and risks that could negatively impact the drone will be reviewed carefully during team meetings so that dependencies, mitigation actions, and the schedule remain up to date.

V. Summary and Conclusions

The Titan Sea Drone conceptual design combines battery-powered flight on a foreign moon, RTG recharging during the night, and sample collection via a vacuum while floating on a Titan sea. Adequate stability and control within the flight vehicle's mass and power limits are depended on by the feasibility of the drone. Once the MCR is approved, the immediate priorities are the following: confirm subsystem interfaces, obtain preliminary mass and rotor data, and develop the parameterized MATLAB hover model of the drone. Performance will be evaluated by simulation, staged testing, and readiness reviews. The simulation, staged testing, and readiness reviews will also identify control limitations before free flight. The testing of the Earth demonstrator drone will follow applicable safety procedures and flight regulations.

VI. References

[1] NASA, “Titan Facts,” *NASA Science* [online], URL: https://science.nasa.gov/saturn/moons/titan/facts/ [retrieved 19 September 2026].

[2] McGee, T. G., Adams, D. S., Hibbard, K. E., Turtle, E. P., Lorenz, R. D., Amzajerdian, F., and Langelaan, J. W., “Guidance, Navigation, and Control for Exploration of Titan with the Dragonfly Rotorcraft Lander,” *2018 AIAA Guidance, Navigation, and Control Conference*, AIAA Paper 2018-1330, Jan. 2018. doi:10.2514/6.2018-1330.